PRACTICAL:3

Aim :

Theory :

Code :

```
% Image Enhancement (Point Processing)

clc;
clear;
close all;

a = imread('/MATLAB Drive/Practical3.jpg');
figure, imshow(a), title('Original Color Image');

b = rgb2gray(a);
figure, imhist(b); title ('image histogram');
figure, imshow(b); title ('grayscale image');

% Histogram Equalization

c = histeq(b);
figure, imshow(c); title ('Histogram Equalized Machine');
figure, imhist(c), title('Histogram Equalized Image');
```

```
%% Read the image

img = im2double(b); % Convert to double for processing

% Apply logarithmic transformation
d = 10; % Scaling constant
log_img = d * log(1 + img);

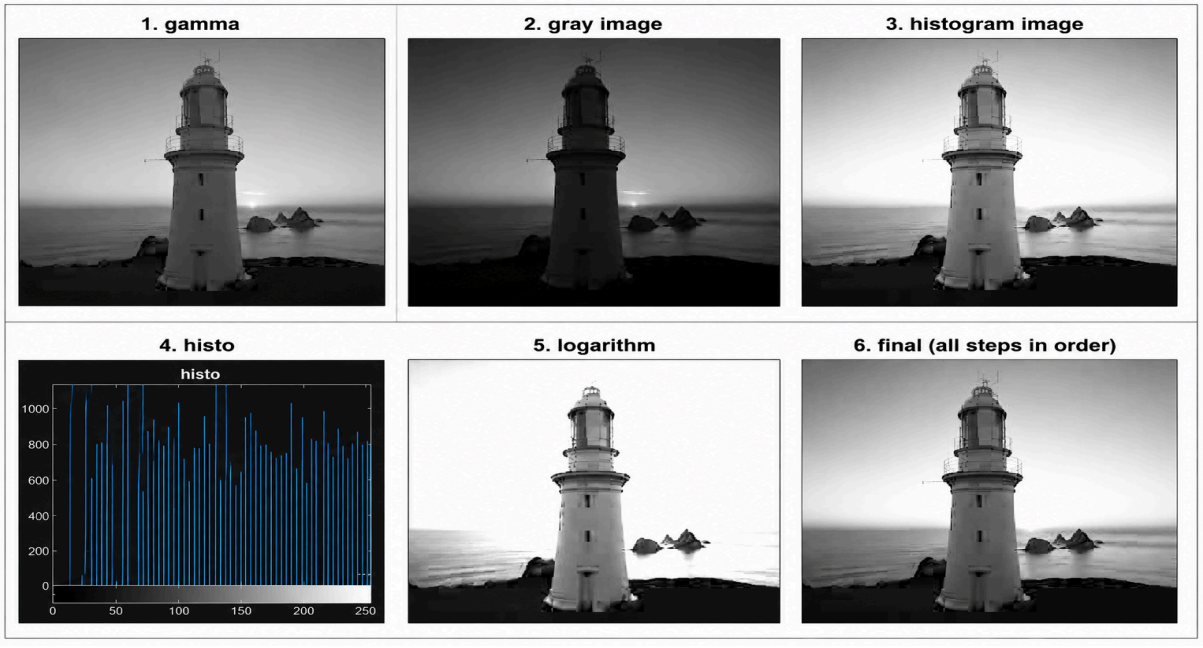
% Display results
figure; imshow(log_img); title('Logarithmic Transformation');


%% Exponential (Power-Law) Transformation

% Apply power-law (gamma) transformation
gamma = 0.5; % Gamma value < 1 brightens the image, > 1 darkens it
c = 1; % Scaling constant
gamma_img = c * (img.^gamma);

% Display results
figure; imshow(gamma_img); title(['Power-Law (Gamma = ', num2str(gamma), ')']);
```

Output:



Conclusion: